

Benchmarking LASSO, PCA, and VAE for Breast Cancer Survival Prediction Using Gene Expression and Clinical Data

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Summary

This project evaluated multiple approaches to predicting distant recurrence-free survival (DRFS) in breast cancer using gene expression and clinical data. Starting with more than 20,000 genes, univariate Cox regression identified several thousand candidates, which were further reduced to a few dozen genes using LASSO-Cox. To benchmark genomic against traditional clinical information for prognostic analysis, we compared model performance across three input sets: (i) clinical variables only; (ii) clinical variables plus gene-expression data; and (iii) gene-expression data only, using both Cox proportional hazards (PH) models and Random Survival Forests (RSF). We also tested dimensionality-reduction methods—principal component analysis (PCA) and variational autoencoder (VAE)—applied to the gene-expression data; the resulting principal components and VAE latent variables were subsequently modeled with Cox regression.

Extensive bootstrap cross-validation was used to evaluate models by mean time-dependent AUC (iAUC) and concordance index (C-index) across iterations. Data preprocessing included Multivariate Imputation by Chained Equations (MICE) for missing clinical values, z-score normalization of gene-expression data, and strict separation of training and validation sets to prevent data leakage. Across 100 bootstrap resamples with replacement, none of the models achieved strong performance on the test set. By comparison, the PCA-based Cox model modestly outperformed the others (mean iAUC = 0.57; mean C-index = 0.56). The single best PCA-Cox model—identified on one train/test split by the highest score—was retained as the final model for future prediction.

Introduction

Breast cancer remains a leading threat to women’s health. Traditional survival analysis in breast cancer has relied primarily on clinicopathologic variables such as tumor stage, histologic grade, lymph-node status, and receptor status. However, even the strongest predictors of metastasis—for example, nodal status and histologic grade—often fail to accurately stratify tumors by their clinical behavior.(Veer et al. 2002) As high-throughput sequencing has become widespread and more affordable, genomic data have become central to predicting outcomes and guiding treatment decisions.(Venetis et al. 2024) The tumor’s molecular profile—specific mutations and gene-expression patterns—drives its behavior, which in turn leads to different clinical trajectories.

Transcriptomic datasets commonly used in survival studies include RNA-seq and microarrays. Standard methods in survival analysis include Kaplan–Meier curves with the log-rank test and multivariable Cox PH models. Regularized Cox models (e.g., LASSO-Cox) and RSF are also widely applied, including for feature selection. Researchers use unsupervised clustering to identify molecular subtypes associated with distinct outcomes. In recent years, deep learning–based survival methods have gained popularity for multimodal data, complementing traditional statistical models.(Wiegreb et al. 2024)

In this project, we built and compared classical survival models—specifically Cox PH models and RSF—and assessed the predictive contribution of genomic features relative to clinical variables. For gene expression data, beyond selecting predictive genes via feature engineering for direct modeling, we applied PCA for dimension reduction and trained a VAE to learn compact latent representations that are subsequently entered

into Cox models. During exploratory analyses, patient samples were clustered by gene expression to define molecular subtypes, and survival differences across subtypes were evaluated.

Data Source

The data for this project were sourced from the internationally renowned NCBI GEO Database (Gene Expression Omnibus) under the accession number GSE7390, which includes a total of 198 breast cancer patients. All patients were lymph node-negative. Clinical variables include age, tumor grade, estrogen receptor status (ER status), and tumor size. The gene expression data were obtained using microarray technology and comprise 22,283 variables. The target variables include: Distant Metastasis-Free Survival Status(e.dmfs): A binary variable indicating whether the patient experienced disease recurrence (yes/no). Time to Distant Metastasis-Free Survival(t.dmfs): The duration from treatment to the occurrence of disease recurrence, measured in days. For patients who did not experience recurrence, the last known recurrence-free time was recorded (i.e., right-censored data).

Table 1: Subset of Gene Expression Matrix (n = 5)

Probe_ID	GSM177885	GSM177886	GSM177887	GSM177888	GSM177889
1007_s_at	10.7808	11.3354	11.0281	11.8477	12.3592
1053_at	8.6742	9.3548	9.0539	9.1399	9.1967
117_at	7.7386	7.7637	6.3276	7.0326	7.8734
121_at	9.2855	9.0258	9.2344	9.6564	9.1742
1255_g_at	6.6105	4.8099	4.6220	5.5894	5.3252

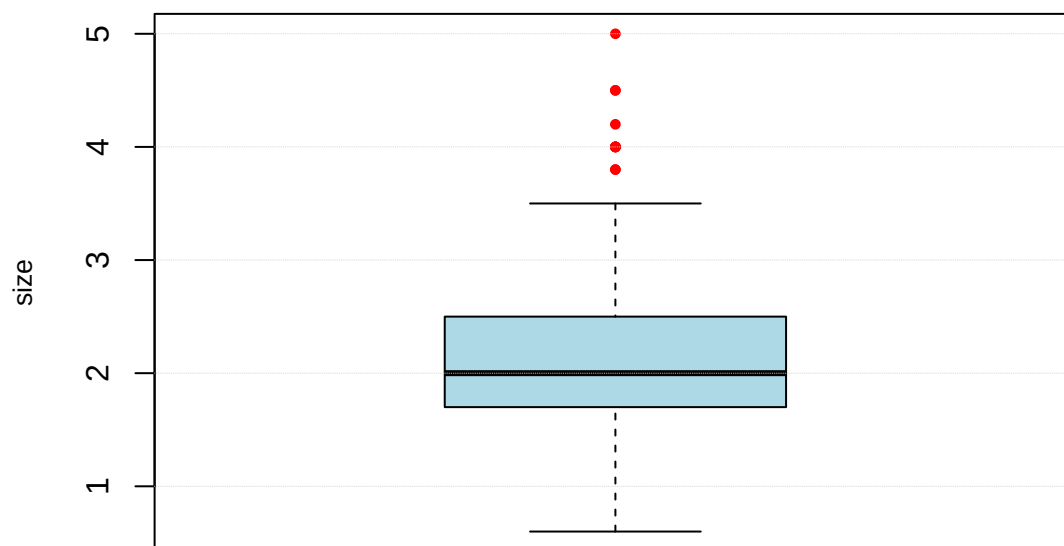
Table 2: Subset of Clinical Data (n = 5)

geo_accession	t_dmfs	e_dmfs	age	er	grade	size	hospital
GSM177885	723	1	57	0	3	3.0	GUY
GSM177886	6591	0	57	1	3	3.0	GUY
GSM177887	524	1	48	0	3	2.5	GUY
GSM177888	6255	1	42	1	3	1.8	GUY
GSM177889	3822	1	46	1	2	3.0	GUY

Data Preprocessing and Exploratory Data Analysis (EDA)

Clean clinical data: handle missing values and outliers

Figure 1: Boxplot of Tumor Variable: size



The box plot of clinical data shows no outliers.

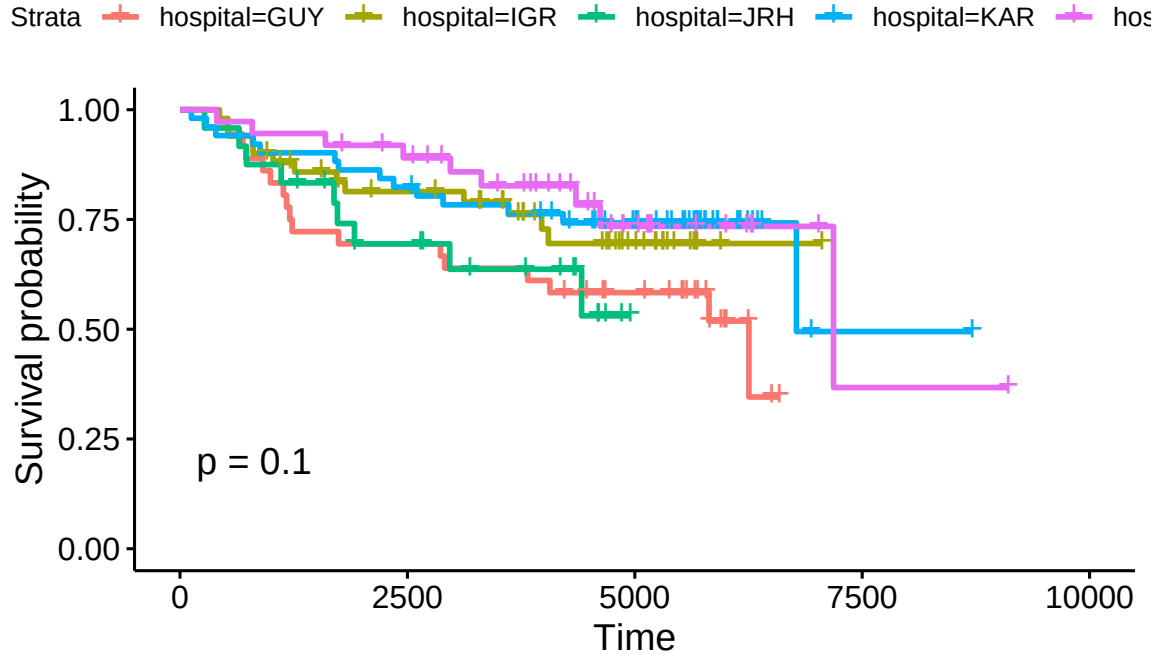
Table 3: NA Audit Report

Column	NA_Count	Class	Mice_Method
grade	2	factors level > 2	polyreg

The clinical dataset exhibited missingness exclusively in the grade column, with two incomplete observations. As grade represents an ordinal factor variable with three levels, multiple imputation was performed using the proportional odds logistic regression (“polr”) method within the MICE framework. The final imputed values were derived through majority voting across the multiply imputed datasets. To prevent data leakage, the MICE algorithm was trained solely on the training subset. The validation set was subsequently imputed using the parameters estimated from the training data, thereby preserving the validation set’s integrity for unbiased model performance assessment.

There are 5 hospitals in the clinical dataset. The Kaplan-Meier survival curves generated by ggsurvplot demonstrate partially overlapping patterns across these hospitals, and the log-rank test yields a p-value of 0.1. While this suggests some degree of heterogeneity in survival outcomes among different hospital sites, the p-value > 0.05 indicates that the observed differences are not statistically significant at the conventional alpha level of 0.05. In view of the limited sample size and the lack of statistical significance, hospital-level random effects were only subjected to a brief test within the subsequent PCA-Cox modeling and were not treated as a primary factor elsewhere.

Figure 2: Kaplan–Meier Survival Curves by Hospital



Exploring the gene data with hierarchical cluster

The value of gene expression ranges from -2.8583908 to 16.9159279. This range is typical data converted by Log2. The summary table below indicates the absence of both outliers and missing values.

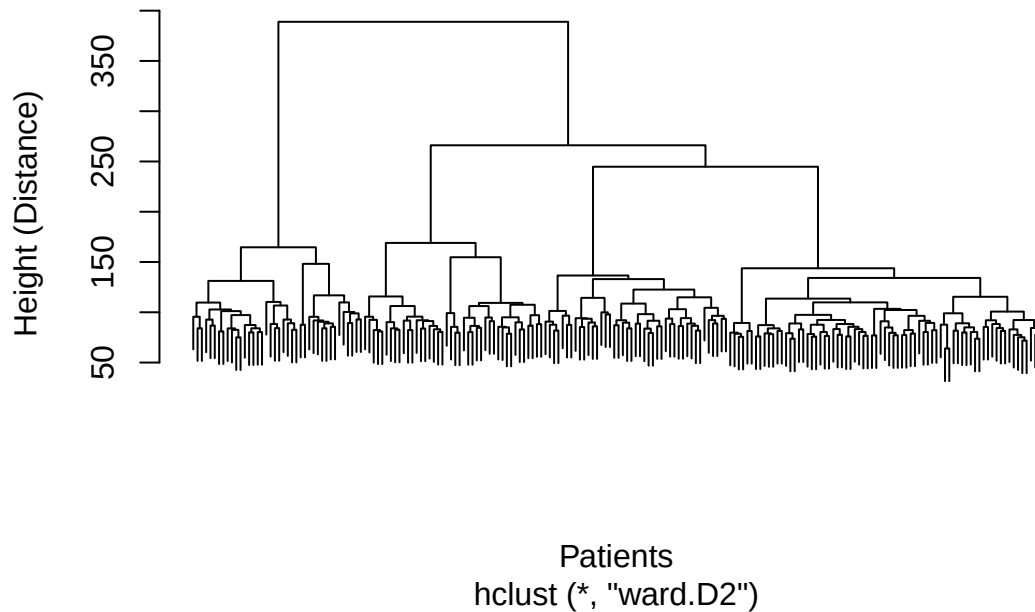
Table 4: Summary of Genomic Data

Statistic	Min.	X1st.Qu.	Median	Mean	X3rd.Qu.	Max.
Mean	0.9949	5.7500	7.6778	7.6224	9.3100	16.1378
SD	0.2178	0.5802	0.8402	0.9094	1.1772	3.9008

Leveraging genomic features allows us to stratify patients into distinct molecular subgroups. We then assess whether survival curves differ meaningfully across these clusters, aiding the identification of molecular determinants that influence prognosis.

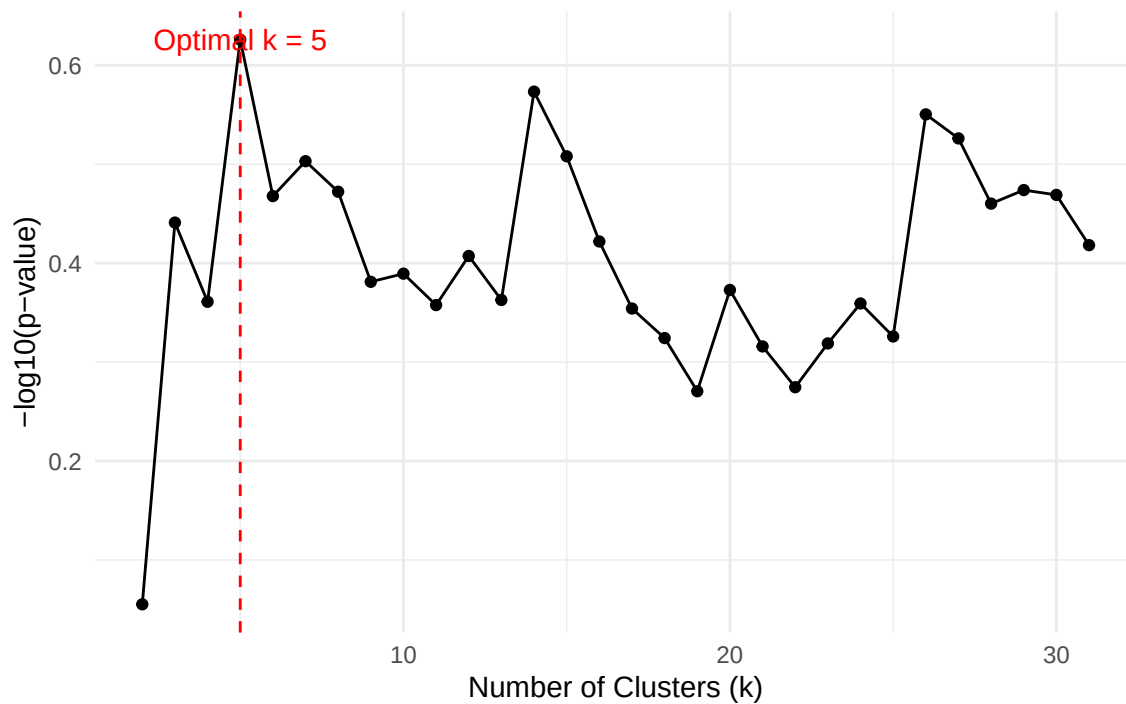
Because of computational constraints, we first performed feature preselection by taking the top 1500 most variable genes (variance ranking). We computed pairwise Euclidean distances to form a dissimilarity matrix and applied hierarchical clustering using Ward's minimum-variance linkage, which promotes compact, low-within-cluster variance groupings.

Figure 3: Hierarchical Clustering Dendrogram of Samples



To determine the optimal clustering resolution, cluster numbers ranging from 2 to 31 were systematically evaluated. For each clustering solution, Kaplan-Meier survival analysis was conducted accompanied by log-rank testing to assess inter-cluster survival differences.

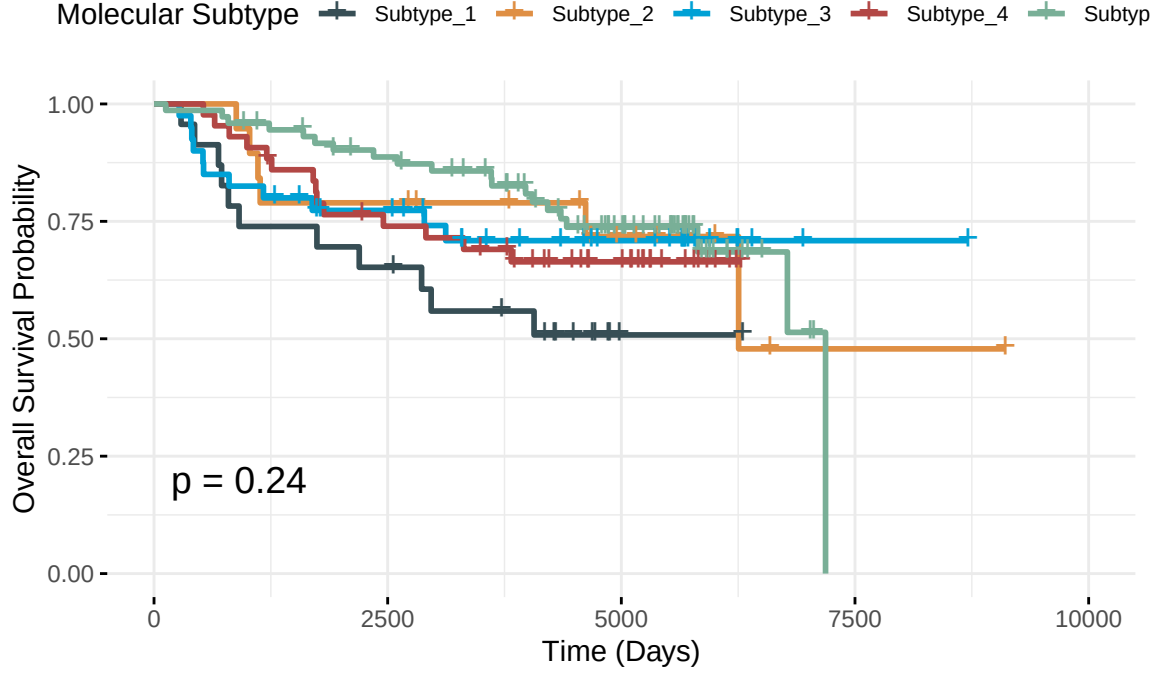
Figure 4: Log-rank Test P-values for Molecular Subtype Clustering



The optimal partition 5 was selected based on the most statistically significant separation of survival curves,

as determined by the minimal log-rank p-value across all tested configurations. Following the division of samples into 5 groups, the KM log-rank test yielded a p-value of 0.236621, indicating no statistically significant differences in survival outcomes among these groups.

Figure 5: Kaplan–Meier Survival Curves by Molecular Subtype



Train Test Splitting and MAD filter on genomic data

Further processing will include: train test data splitting, clinical miss data imputation, genetic feature pre-selection, and standardization of genomic data.

Trough resampling with replacement, we performed train–test splits and applied multiple imputation to the clinical covariates. For the genomic features, we applied median absolute deviation (MAD) filtering on the training set to remove genes with minimal expression variability. Genes in the top 0.75 quantile of variability were retained for downstream analysis. After preprocessing, the number of gene features was reduced from 22283 to 16712. The validation set was restricted to the same gene subset to ensure analytical consistency and prevent data leakage.

COX PH Regression

Univariate Cox PH Regression

We performed univariate Cox PH regression to screen for prognostic genes. The multivariable cox PH model is a semi-parametric model that relates survival time to multiple predictors. Its specification is:

$$h(t|X) = h_0(t) \times \exp(\beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p)$$

where: - $h(t|X)$ is the hazard function at time t given covariates X - $h_0(t)$ is the baseline hazard function - $\exp(\beta_i X_i)$ represents the hazard ratio for each covariate

The univariate cox PH model just has one predictor.

We fit the univariate Cox PH models for each gene. Genes showing a significant association (Wald z-test, < 0.05) were identified and compiled into a results table. The number of gene features was further reduced to 2619.

Table 5: Univariate Cox Significant Genes (n = top 5)

gene	coef	HR	SE	p.value
202859_x_at	0.4874	1.6281	0.0831	0
200760_s_at	-1.0684	0.3436	0.2072	0
214210_at	-0.8528	0.4262	0.1659	0
220106_at	0.5103	1.6658	0.0995	0
205231_s_at	-0.8915	0.4100	0.1767	0

LASSO Cox PH Regression

To further refine the feature set, we applied LASSO (Least Absolute Shrinkage and Selection Operator) -penalized Cox PH regression. The L1 penalty enables simultaneous coefficient shrinkage and variable selection, yielding a parsimonious subset of prognostic genes, with the penalty strength tuned during model training. The objective function combines partial likelihood with L1 penalty:

$$\min_{\beta} \left[-\frac{1}{n} \ell(\beta) + \lambda \sum_{j=1}^p |\beta_j| \right]$$

where:

- $\ell(\beta)$ = partial likelihood of the Cox model
- λ = regularization parameter controlling shrinkage
- $\sum_{j=1}^p |\beta_j|$ = L1 penalty term

For the multivariable or Lasso Cox survival modeling it is essential to compare effect sizes (hazard ratios) on a common scale. Therefore, Z-score normalization was applied individually to each gene and numerical clinical variable. Additional feature selection was performed using `remove_high_corr_genes` function with a correlation coefficient cutoff of 0.90 to address multicollinearity among predictors. To similarly prevent information leakage, the validation set was standardized utilizing the mean and standard deviation parameters calculated solely from the training dataset.

Table 6: Genomic test set (n = 5), scaled (training-set parameters)

gene	GSM177885	GSM177887	GSM177891	GSM177894	GSM177895
1255_g_at	1.4685	-0.1009	0.5835	0.7337	-0.5005
1294_at	-0.9424	-1.1126	-1.9966	-0.7195	-0.8035
1729_at	1.3576	-0.7607	0.4220	-1.6236	0.7261
200074_s_at	-0.2605	0.6378	-0.6959	0.4725	0.2196
200081_s_at	-1.5481	-0.3198	-0.7717	0.0628	0.2622

The LASSO Cox regression procedure incorporated repeated cross-validation to enhance the robustness of gene selection across multiple iterations. The resulting genes were prioritized based on selection frequency and mean coefficient magnitude, with a retention threshold set at ≥ 0.8 frequency.

Regarding sample size considerations, (Peduzzi et al. 1995) initially proposed a conservative requirement of 10 events per variable (EPV) for Cox regression. (Vittinghoff and McCulloch 2007) later demonstrated that 5–9 EPV may be sufficient for many analytical scenarios. Consistent with the latter guidelines and

considering the cohort size limitations, an EPV threshold of 5 was implemented for additional feature selection. Since there were 69 events in the training data, the maximum number of variables allowed is 13 . Consequently, 13 genes were ultimately selected as feature variables.

However to assess whether adding gene-expression features improves the discriminative performance of Cox models over clinical covariates, we appended the genomic features directly to the clinical set without enforcing the events-per-variable (EPV) rule.

Table 7: Multi_Iteration_LASSO_Genes_Selection_Frequency_0.8_EPV_5

	gene	freq	mean_coef
202859_x_at	202859_x_at	1	0.5981
220106_at	220106_at	1	0.3690
216168_at	216168_at	1	0.3173
204531_s_at	204531_s_at	1	0.2824
217263_x_at	217263_x_at	1	-0.1861
200760_s_at	200760_s_at	1	-0.1856
215502_at	215502_at	1	-0.1679
214013_s_at	214013_s_at	1	-0.1378
218054_s_at	218054_s_at	1	-0.1208
215969_at	215969_at	1	0.1163
215348_at	215348_at	1	0.1122
214210_at	214210_at	1	-0.1074
201815_s_at	201815_s_at	1	-0.1050

Risk score calculation and Kplan Maier analysis between groups

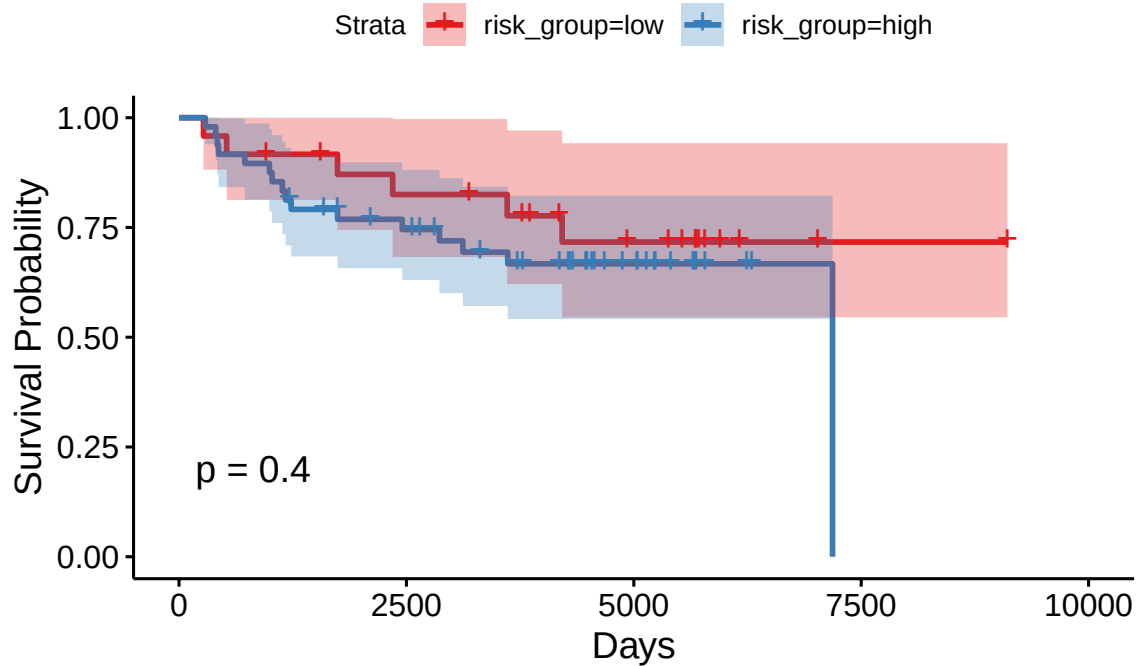
The risk score for patient was calculated as:

$$\text{risk score} = \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_i x_i + \dots + \beta_p x_p$$

where: - β_i represents the coefficient for each variable - x_i represents the value of each predictor variable - p is the total number of variables in the model

Subsequently, patients were stratified into high-risk and low-risk groups based on the median risk score of the training cohort. This same median value, derived from the training set, was then applied as the cutoff to classify patients in the validation set into high- and low-risk groups following the calculation of their risk scores.

Figure 6: Kaplan–Meier Survival Curves by Risk Group



The Kaplan–Meier survival curves show substantial overlap between the two groups, and the log-rank test confirms no statistically significant difference in survival ($p = 0.399324$). This indicates that stratifying patients by risk score does not yield a clear survival separation between groups.

Safeguards Against Separation in Cox Models

In certain training datasets, Cox models can exhibit (quasi-)complete separation, causing coefficients to diverge and the concordance index (C-index) to appear spuriously high. To screen out such pathological fits, we enforce two checks: if the model's C-index exceeds `min_concord` (e.g., 0.97) or any coefficient's absolute value exceeds `coef_max` (e.g., 10), the model is rejected. The following shows the model's training results from one random train-test split.

As seen in the table 8 the Cox model shows that Grade2 is significant: `coef 0.87` ($p = 0.034$) vs. the reference; grade3 is not significant. Several genes show strong associations (e.g., `ge_202859_x_at` `coef 1.39`; `ge_220106_at` `coef 1.23`; `ge_217263_x_at` protective, `coef -1.07`). Age, ER, tumor size are not statistically significant. The coefficient for tumor size is negative, which is counterintuitive. This may be due to limited data size, abnormal data distribution, or the influence of confounding factors. Overall fit tests (LR/Wald/Score) are highly significant. The PH assumption is violated globally ($p = 1e-12$). The validation of the model is presented following the section on Model Discrimination Assessment Metrics. After identifying significant genes, we will fit two models—a Cox PH model and a Random Survival Forest—with training/validation conducted under bootstrap resampling.

Random Survival Forest model

Random Survival Forest (RSF) is an extension of the Random Forest method to the survival analysis domain. It consists of many survival trees, built using bootstrap samples and random feature selection at each split. Splitting rule: Instead of maximizing information gain (e.g., Gini index or entropy in classification), survival trees use the log-rank test to find the feature and split point that best separate

Table 8: Full Cox Model Analysis Summary

	variable	coef	zscore	pvalue
age	age	0.1362	0.8352	0.4036
er1	er1	-0.2095	-0.5314	0.5951
grade2	grade2	0.8709	2.1251	0.0336
grade3	grade3	0.4722	0.8791	0.3794
size	size	-0.2921	-1.6147	0.1064
ge_202859_x_at	ge_202859_x_at	1.3924	5.8384	0.0000
ge_220106_at	ge_220106_at	1.2306	5.8280	0.0000
ge_216168_at	ge_216168_at	1.0747	5.3368	0.0000
ge_204531_s_at	ge_204531_s_at	1.2796	5.7956	0.0000
ge_217263_x_at	ge_217263_x_at	-1.0729	-5.0881	0.0000
ge_200760_s_at	ge_200760_s_at	-0.2386	-1.3325	0.1827
ge_215502_at	ge_215502_at	-0.6351	-2.7458	0.0060
ge_214013_s_at	ge_214013_s_at	-0.5531	-2.8845	0.0039
ge_218054_s_at	ge_218054_s_at	-0.6223	-3.1623	0.0016
ge_215969_at	ge_215969_at	0.8210	4.3256	0.0000
ge_215348_at	ge_215348_at	0.5922	2.8774	0.0040
ge_214210_at	ge_214210_at	-0.8360	-4.0875	0.0000
ge_201815_s_at	ge_201815_s_at	-0.2712	-1.7454	0.0809

Table 9: Full Cox Model Proportional Hazards Test

	chisq	df	p
age	8.1376	1	0.0043
er	2.7789	1	0.0955
grade	23.4547	2	0.0000
size	5.0571	1	0.0245
ge_202859_x_at	0.2619	1	0.6088
ge_220106_at	0.4416	1	0.5063
ge_216168_at	0.4366	1	0.5088
ge_204531_s_at	1.2373	1	0.2660
ge_217263_x_at	14.9550	1	0.0001
ge_200760_s_at	4.6958	1	0.0302
ge_215502_at	13.2689	1	0.0003
ge_214013_s_at	11.4337	1	0.0007
ge_218054_s_at	0.6212	1	0.4306
ge_215969_at	10.9696	1	0.0009
ge_215348_at	5.8137	1	0.0159
ge_214210_at	1.4410	1	0.2300
ge_201815_s_at	0.3166	1	0.5737
GLOBAL	96.2649	18	0.0000

survival outcomes. For standard random forests, the leaf output is class label (classification) or numeric value (regression). In RSF, each terminal node (leaf) stores an estimated cumulative hazard function (CHF) ($H(t)$). For a new sample, each tree returns the CHF of the leaf the sample falls into, and the ensemble prediction is the average CHF across trees $H_{\text{avg}}(t)$ and convert to a survival function:

$$S(t) = \exp(-H_{\text{avg}}(t))$$

RSF outputs a survival curve, which gives the estimated probability of surviving beyond any time point (t). Besides individual survival prediction, RSF also supports variable importance analysis. This analysis

utilized a random survival forest comprising 1,000 trees, with model training conducted through k-fold cross-validation, followed by selection of the optimal model for testing evaluation. Variable importance was determined using the full dataset Table 11.

Table 10: Variable Importance from Random Survival Forest model

	Variable	Importance
ge_202859_x_at	ge_202859_x_at	0.1790
ge_204531_s_at	ge_204531_s_at	0.1415
ge_220106_at	ge_220106_at	0.1026
ge_215969_at	ge_215969_at	0.0987
ge_200760_s_at	ge_200760_s_at	0.0948
ge_215502_at	ge_215502_at	0.0923
ge_217263_x_at	ge_217263_x_at	0.0917
ge_214210_at	ge_214210_at	0.0643
ge_218054_s_at	ge_218054_s_at	0.0572
ge_201815_s_at	ge_201815_s_at	0.0564
ge_216168_at	ge_216168_at	0.0510
ge_215348_at	ge_215348_at	0.0272
ge_214013_s_at	ge_214013_s_at	0.0259
age	age	0.0244
size	size	0.0074
grade	grade	0.0013
er	er	0.0010

Model Discrimination Assessment Metrics

C-index (Concordance Index)

In this study, the discriminatory power of the survival model was assessed using the average C-index (Concordance Index) and average iAUC (Integrated Area Under the Curve) obtained through repeated cross-validation. The C-index is calculated by randomly selecting a pair of individuals (i, j) . The pair is considered concordant if: - Individual i has an actual event time shorter than individual j , and - The model assigns a higher risk score (or shorter predicted survival time) to individual i . The C-index represents the proportion of concordant pairs among all comparable pairs:

$$C = \frac{\text{Number of concordant pairs}}{\text{Number of comparable pairs}}$$

Interpretation of Values:

- $C = 0.5$ Prediction is no better than random guessing
- $C = 1.0$ Perfect discriminatory ability
- $0.7 \leq C < 0.8$ Acceptable discrimination
- $0.8 \leq C < 0.9$ Excellent discrimination
- $C \geq 0.9$ Outstanding discrimination

iAUC (Integrated Area Under the Curve)

The iAUC is the integrated form of time-dependent AUC. It measures a model's discriminatory ability at different time points and provides an overall summary score through integration (weighted averaging) over

time. The calculation proceeds in two main steps:

- i. Compute time-dependent AUC (t): At a specific time point t , define two groups:
 - The event group: individuals who experienced the event by time t .
 - The non-event/survival group: individuals who remain event-free beyond time t (including those censored after t).

Calculate the standard AUC (Area Under the ROC Curve) for classifying these two groups at time t based on the model's predicted risk scores. Since the risk scores are fixed, at each time point t we simply reclassify the samples into the event group and the survival group. We then compare the risk scores to see whether the model correctly ranks those who experienced the event before t ahead of those who survived beyond t .

- ii. Integrate over all observed time points: The overall iAUC is obtained by integrating the time-dependent AUC(t) function over the entire follow-up period $[0, \tau]$:

$$\text{iAUC} = \int_0^{\tau} \text{AUC}(t) w(t) dt$$

Where: - τ is the maximum observation time. - $w(t)$ is a weight function, often based on the distribution of event times or the size of the risk set at time t to account for the varying precision of the estimate across time.

Interpretation of iAUC Values is same as the C-index.

Both the C-index and iAUC provide valuable insights into model performance: The C-index primarily assesses risk ordering but does not account for time-dependent dynamics in model performance which is addressed by the iAUC. The combination of these metrics through repeated cross-validation ensures a robust assessment of the survival model's discriminatory power. The table 12 shows the C-index and iAUC of the COX and RSF models, which were built using both genetic and clinical data, along with the performance of models using clinical data alone, all evaluated on the test set.

Table 11: Optimal Iteration Performance in 100 Random Train-Test Splits

Model	iAUC	C_Index	Sum_iAUC_Cindex
COX Clinical-Genomic	0.7452	0.7257	1.4709
RSF Clinical-Genomic	0.7734	0.7082	1.4816
COX Clinical-only	0.7623	0.7537	1.5160

On the validation data set, the Time-weighted average AUC (iAUC) is 0.7451841 and the Concordance Index (C-index) is 0.7257. However, the above results come from a single random train-test split; a comprehensive assessment requires cross-validation.

Botstrap cross validation using three separate feature sets

To compare the performances of the Cox PH model and the Random Survival Forest model, each algorithm were trained and evaluated using three separate feature sets: an exclusive clinical variable set, an exclusive gene expression variable set, and a combined set integrating both clinical and genomic features. Through bootstrap cross validations with replacement, we calculate the everage of C_index and iAUC.

Across 100 bootstrap resamples, the Clinical-Genomic Integrated Cox model yielded a mean iAUC and C-index with standard deviations around 0.09 indicating no performance advantage over a random guess. Similarly, the other models exhibited comparable performance. Below, we proceed with dimensionality reduction of the gene expression data followed by modeling using the Cox PH approach.

Table 12: 3 Model Performance Comparison from 100 Bootstrap Iterations

Model_Type	MCoxiAUC	SdCiAUC	MCoxCinx	SdCCinx	MRSFiAUC	SdRiAUC	MRSFCinx	SdRCinx
Clinical-only	0.55	0.08	0.52	0.09	0.52	0.09	0.50	0.09
Genomic-only	0.52	0.10	0.52	0.10	0.52	0.09	0.51	0.09
Clinical-Genomic Integrated	0.54	0.10	0.54	0.10	0.52	0.09	0.50	0.09

Table 13: Abbreviations for Survival Analysis Performance Metrics

Abbreviation	Full_Name
MCoxiAUC	Mean Cox iAUC
SdCiAUC	sd of Cox iAUC
MCoxCinx	Mean Cox C-index
SdCCinx	sd of Cox C-index
MRSFiAUC	Mean RSF iAUC
SdRiAUC	sd of RSF iAUC
MRSFCinx	Mean RSF C-index
SdRSFCinx	sd of RSF C-index

PCA Cox: Cox model built on Principal Component Analysis (PCA) results

PCA is an unsupervised, linear method that rotates the data to new, orthogonal axes (principal components) that capture the largest possible variance first. Each sample gets scores on these components; each gene has loadings that show how strongly it contributes. PCA is typically computed by an eigen-decomposition of the covariance (or correlation) matrix, or via singular value decomposition (SVD).

PCA reduces dimensionality, mitigates multicollinearity, and can denoise high-dimensional data. In gene expression studies, PCA can compress ~20,000 genes into a few dozen components that retain most variation, yielding more stable estimates and faster computation in downstream analyses.

In this study, we reduced the dimensionality of gene-expression data via PCA fitted on the training set and used the resulting principal component (PC) scores as predictors in Cox PH model. The validation set was projected using the training-set centering, scaling, and rotation (loadings). Model performance was summarized over 100 bootstrap resamples. When selecting the number of PCs, we enforced an events-per-variable (EPV) rule of 3. The goal is to ensure that the retained principal components explain at least 50% of the variance in the original data.

Table 14: Cox model built on PCA CV result (n_runs = 100)

Metric	Mean	SD
C-index	0.57	0.11
CoxMe C-index	0.42	0.07
iAUC	0.56	0.12
Total Score	1.13	0.22

Exploratory analyses suggested between-hospital heterogeneity in survival ($p = 0.10$). Motivated by this, we also fit a Cox mixed-effects model (coxme) with a hospital-level random effect, but it did not improve validation performance. The mean C-index performance fell below random chance levels. This sub-random performance is presumably attributable to severe sample insufficiency after incorporating random effects into the model.

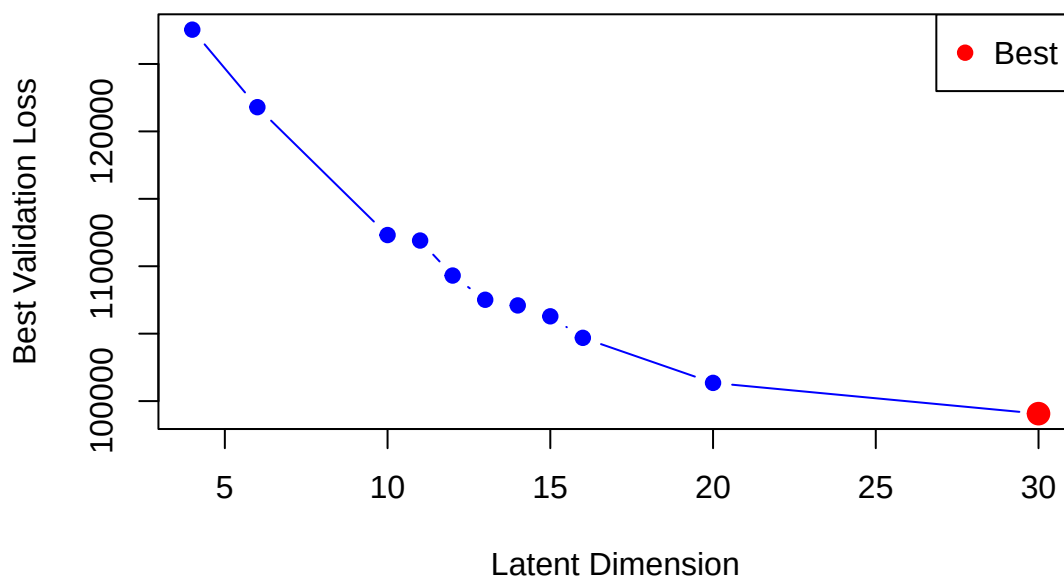
VAE Cox: Cox model built on VAE latent representation

A Variational Autoencoder (VAE) is a type of deep generative model that learns to represent high-dimensional data, such as gene expression profiles, in a lower-dimensional latent space. In this project this method was

also explored, with the resulting latent variables incorporated into Cox regression modeling.

VAE consists of two parts: an encoder, which compresses the input data into a set of latent variables, and a decoder, which reconstructs the data from these variables. Unlike a standard autoencoder, a VAE assumes a probabilistic distribution for the latent variables (typically Gaussian), allowing it to generate new data samples by sampling from this distribution. This property makes VAEs useful for capturing complex patterns in the data and creating smooth latent representations that can be used in downstream analyses, such as survival modeling.

Figure 7: VAE Performance Across Latent Dimensions



In this project, VAE models were first trained on the training dataset across multiple latent dimensions 4, 6, 10, 11, 12, 13, 14, 15, 16, 20, 30, and the reconstruction loss + Kullback–Leibler (KL) divergence was evaluated on an independent test set. The latent dimension 30, which produced the lowest validation loss, was compared with the events-per-variable (EPV) rule, and the smaller value was selected as the basis for subsequent Cox regression modeling.

A VAE model was first constructed using the training data, from which the latent representations of training data were extracted. A Cox model based solely on these latent variables was then built. The validation data were subsequently transformed into latent representations with the VAE model and used to evaluate the Cox model. Fifty iterations of cross-validation with replacement were conducted.

It is worth noting that, to reduce computational load in the VAE pipeline, we preselected genes prior to training by (i) applying median absolute deviation (MAD) filtering to remove low-variability genes, (ii) conducting univariate Cox screening, and (iii) pruning highly correlated genes. We then standardized the training set (with the same centering/scaling applied to validation data). These preprocessing steps were not applied in the PCA-based workflow.

Based on the test results, none of the models demonstrate satisfactory discriminative performance on the test set. Nevertheless, since the project requires selecting one model as the predictive model for future use, we have chosen the Cox model constructed after applying PCA as the predictive model because it has a slight better performance.

Table 15: 5 Model Performance Comparision

Model_Type	MCoxiAUC	SdCiAUC	MCoxCinx	SdCCCinx	MRSFiAUC	SdRiAUC	MRSFCinx	SdRCinx
Clinical-only	0.55	0.08	0.52	0.09	0.52	0.09	0.50	0.09
Genomic-only	0.52	0.10	0.52	0.10	0.52	0.09	0.51	0.09
Clinical-Genomic Integrated	0.54	0.10	0.54	0.10	0.52	0.09	0.50	0.09
PCA-Genomic-only	0.56	0.12	0.57	0.11	NA	NA	NA	NA
VAE-Genomic-only	0.54	0.10	0.54	0.10	NA	NA	NA	NA

Model application and Prediction

In order to perform personalized survival prediction for new patients using a pre-trained PCA-COX model, the genomic data is needed to be standardized using the same center and scale parameters from the original training set to maintain consistency. Then the standardized data is transformed into principal component space using the rotation matrix from the trained PCA model, retaining only the most significant components based on the original variance threshold.

The pre-trained COX model generates three key predictions: a linear risk score indicating relative risk, a hazard ratio representing the multiplicative increase in risk compared to the baseline, and a personalized survival probability estimate specifically at the patient’s own observed follow-up time.

Table 16: PCA Cox prediction VS real survive time

	risk_score	hazard_ratio	survival_P_at_val_time	val_time_dmfs	val_event_dmfs
GSM178024	-0.79	0.46	0.83	6135	0

Discussion

We evaluated multiple Cox PH models and RSF on clinical and gene-expression data, and also built Cox models on PCA and VAE-derived representations. All models performed suboptimally. Likely contributors include: (1) limited sample size and few events—low events-per-variable (EPV) leading to unstable/overfit estimates, particularly as nested cross-validation further reduces the training data size; (2) violations of the PH assumption; (3) the $p \gg n$ regime and multicollinearity in gene data, which harm stability and generalization; (4) suboptimal hyperparameters (e.g., for VAE and RSF), suggesting a grid search over latent_dim, hidden_dim, ntree, etc.; and (5) weak signal—clinical/genomic predictors may have limited prognostic value, motivating alternative biological stratifications or data types.

For follow-up, one could fit RSF on PCA scores and on VAE latent variables. The relatively better performance of PCA may reflect preprocessing choices—for example, the absence of univariate Cox/LASSO pre-screening and/or omission of scaling before PCA in some configurations. These steps warrant careful reconsideration to avoid degrading signal representation or injecting unnecessary noise.

Conclusion

Genomic data are increasingly important for cancer prognosis. Using gene-expression and clinical data from the GEO database (GSE7390), we conducted a survival analysis of breast cancer and compared Cox proportional hazards (PH) models with Random Survival Forests (RSF) across different feature representations. Missing clinical values were handled via multiple imputation by chained equations (MICE). To address high dimensionality in the genomic data, we explored PCA and VAE for dimensionality reduction, L1-penalized Cox, median absolute deviation (MAD) filtering, and correlation pruning. All preprocessing and feature selection were fit on the training set and then applied to the validation set to prevent data leakage. Model performance was evaluated using the C-index and the integrated time-dependent AUC (iAUC). Training and testing employed repeated bootstrap resampling to ensure robust, less-biased estimates. The model ultimately chosen for prospective use was a Cox PH model built on PCA-derived principal components.

All models failed to achieve a clearly discriminative level in validation performance. However, the Cox PH model based on PCA-processed gene expression data slightly outperformed other models on the validation set, despite exhibiting higher standard deviation in its performance metrics. Simply adding gene expression data to clinical data did not improve the performance of the traditional Cox PH model. Additionally, the traditional Cox PH model consistently outperformed the Random Survival Forest across all feature sets. Although validation performance did not reach a clearly discriminative level, the study provides a useful baseline and guidance for future refinement.

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